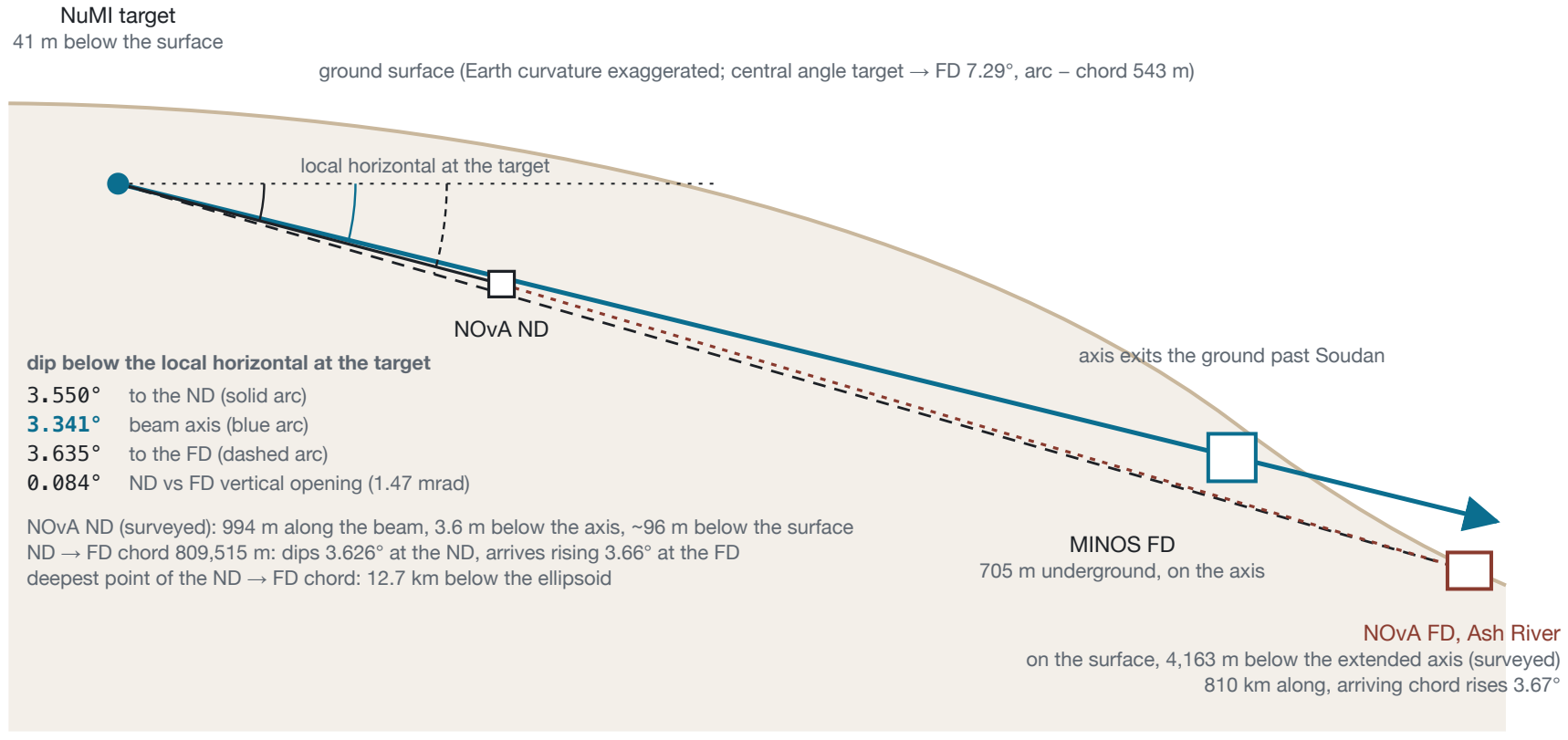
NuMI target
41 m below the surface
ground surface (Earth curvature exaggerated; central angle target → FD 7.29°, arc − chord 543 m)
local horizontal at the target
NOvA ND
dip below the local horizontal at the target
3.550° to the ND (solid arc)
3.341° beam axis (blue arc)
3.635° to the FD (dashed arc)
0.084° ND vs FD vertical opening (1.47 mrad)
NOvA ND (surveyed): 994 m along the beam, 3.6 m below the axis, ~96 m below the surface
ND → FD chord 809,515 m: dips 3.626° at the ND, arrives rising 3.66° at the FD
deepest point of the ND → FD chord: 12.7 km below the ellipsoid
axis exits the ground past Soudan
MINOS FD
705 m underground, on the axis
NOvA FD, Ash River
on the surface, 4,163 m below the extended axis (surveyed)
810 km along, arriving chord rises 3.67°